

3.2 T1, T2 & PROTON DENSITY (PD) WEIGHTING

UNDERSTANDING CONTRAST BY CONTROLLING TR AND TE

By choosing the right TR (repetition time) and TE (echo time), the technologist emphasizes T1 recovery, T2 decay, or Proton Density to create the desired contrast.



KEY CONCEPT

- TR mainly controls T1 weighting
- TE mainly controls T2 weighting
- PD images use long TR and short TE
- "Bright" and "Dark" are RELATIVE to the sequence used

- ↑ Increase / More
- ↓ Decrease / Less
- Minimal Effect

1 WHAT DETERMINES SIGNAL?

MR Signal depends on:



T1 RECOVERY

How fast spins recover longitudinal magnetization



T2 DECAY

How fast spins lose coherence in the transverse plane



PROTON DENSITY (PD)

Number of hydrogen spins in the voxel (intrinsic to the tissue)

Sequence parameters (TR, TE) decide which factor dominates the image contrast.

2 GENERAL EFFECTS OF TR AND TE

EFFECT OF TR (with TE short)

TR Decrease ↓ Increase

Less time for T1 recovery
→ More T1 weighting
(Tissues with short T1 stay brighter)
→ Longer scan

TE More time for T1 recovery
→ Less T1 weighting
→ More PD (if TE short)
→ Shorter scan

EFFECT OF TE (with TR long)

Decrease ↑ Increase

Less time for T2 decay
→ Less T2 weighting
→ More PD (if TR long)
→ Shorter scan

More time for T2 decay
→ More T2 weighting
(Tissues with long T2 stay brighter)
→ Longer scan

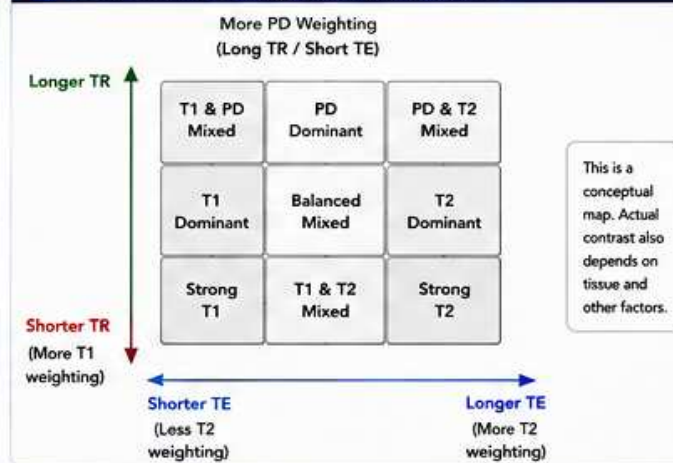
TR controls T1 weighting. TE controls T2 weighting.
PD weighting = Long TR + Short TE.

3 TISSUE CHARACTERISTICS (Typical 1.5T Values)

Tissue	T1 (ms)	T2 (ms)	PD (Relative)
Cerebrospinal Fluid (CSF)	2500–4000	2000–2500	High
White Matter	600–900	80–100	Med-High
Gray Matter	900–1300	100–120	Med
Fat	250–400	60–80	High
Muscle	800–1200	40–60	Med
Blood (flowing)	1200–2000	150–250	High
Cortical Bone	250–400	10–20	Low
Air	—	—	Very Low

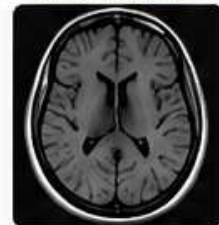
★ Values vary with field strength, temperature, and sequence.

4 HOW CONTRAST CHANGES WITH TR AND TE



5 IMAGE APPEARANCE EXAMPLES (Axial Brain, 1.5T)

A. T1-WEIGHTED
Short TR / Short TE



Typical: TR 500–700 ms
TE 10–20 ms

B. T2-WEIGHTED
Long TR / Long TE



Typical: TR 2500–4000 ms
TE 80–120 ms

C. PROTON DENSITY (PD)-WEIGHTED
Long TR / Short TE



Typical: TR 2000–4000 ms
TE 10–20 ms

Relative Appearance Guide

Tissue	T1	T2	PD
CSF	Dark	Bright	Bright
White Matter	Bright	Gray	Slightly Bright
Gray Matter	Gray	Gray-White	Gray
Fat	Bright	Bright	Bright
Muscle	Gray	Dark-Gray	Gray
Bone	Dark	Dark	Dark
Air	Dark	Dark	Dark

! "Bright" and "Dark" are relative to the sequence. The same tissue can look different depending on TR and TE.

6 HOW TO THINK ABOUT IT



T1 WEIGHTING (short TR, short TE)

- Highlights tissues with short T1 (fat, white matter)
- Good for anatomy and detecting fat, subacute blood
- CSF appears dark



T2 WEIGHTING (long TR, long TE)

- Highlights tissues with long T2 (fluid, edema)
- Good for pathology (edema, demyelination, tumors)
- CSF appears bright



PD WEIGHTING (long TR, short TE)

- Shows relative proton density (cellularity, structure)
- Good for anatomy without strong T1 or T2 effects
- CSF appears bright (slightly less than T2)

7 PARAMETERS THAT MATTER MOST

Goal	Primary	Secondary	Others
T1 Weighting	↓ TR	↓ TE	Flip angle
T2 Weighting	↑ TE	↑ TR	Fat sat / STIR
PD Weighting	↑ TR	↓ TE	—

Typical Ranges (1.5T Brain)

T1	TR 500–700 ms	TE 10–20 ms
T2	TR 2500–4000 ms	TE 80–120 ms
PD	TR 2000–4000 ms	TE 10–20 ms

8 WHAT CAN GO WRONG?

Problem	What You See	Likely Cause (Engineering View)	What to Check
Incorrect contrast (e.g., T1 looks like PD)	Unexpected brightness pattern	Wrong TR/TE, wrong sequence, protocol parameter changed	Verify sequence prescription, TR/TE values, protocol
CSF not dark on T1 or not bright on T2	CSF appears wrong	TR/TE incorrect, T1 used (FLAIR/STIR), saturation band	Check TR/TE, inversion time, sat bands
Low overall signal	All tissues look dark	Low NEX, incorrect coil, bandwidth too high, gain/receiver issue	Check coil, NEX, BW, preamp, cables, receiver gain
Strange banding or shading	Uneven brightness	B0 inhomogeneity, poor shim, fat sat issues	Check shim, B0 map, fat sat frequency

9 ENGINEER CHECKLIST

- ✓ Is the protocol matching the clinical goal (T1, T2, PD)?
- ✓ Are TR and TE values in the expected range?
- ✓ Is FOV and matrix appropriate for the anatomy?
- ✓ Is phase direction optimal (A → P vs P → A)?
- ✓ Are bandwidth and NEX appropriate?
- ✓ Is the correct coil selected and functioning well?
- ✓ Are shimming results acceptable (B0 map)?
- ✓ Are saturation bands / fat sat frequency placed correctly?
- ✓ Do artifacts suggest motion, flow, B0, or RF issues?
- ✓ Repeat test or sequence to confirm if needed.

Quick Reference Summary

Weighting	TR	TE	Key Effect
T1	Short	Short	More T1 (fat, WM bright; CSF dark)
T2	Long	Long	More T2 (fluid/edema bright; CSF bright)
PD	Long	Short	More PD (anatomy without strong T1 or T2)

10 PRACTICAL TIPS

- Always verify TR and TE after protocol changes.
- Look at CSF: dark on T1, bright on T2/PD.
- Compare with prior images and other sequences.
- When in doubt, check TR/TE, T1, fat sat, and coil.
- Small parameter changes can greatly affect contrast!



REMEMBER

TR controls T1 weighting.
TE controls T2 weighting.
PD is achieved with Long TR and Short TE.
Choose the right balance to answer the clinical question.



KEY TAKEAWAY

T1, T2, and PD are the foundation of MRI contrast. Understanding how TR and TE affect these factors helps you recognize images, troubleshoot problems, and support the technologist.